Middle East Technical University
Department of Electrical and Electronics Engineering

EE201 – Circuit Theory I

Lecture 11: Node Voltage Analysis

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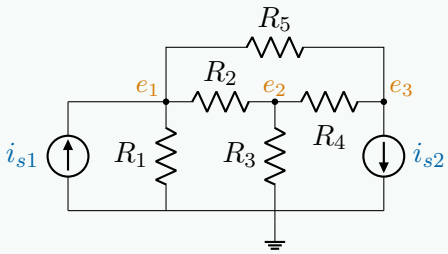
Colour code used in these notes and in the slides		
current i	voltage v, node voltage e	supernode, nodes Ⓢ

Motivation. So far, circuits were solved by simplifying them. Simplifying fundamentally changes the circuit; that is its whole point. To find the quantities of the original circuit, we must go back step by step, and some circuits cannot be simplified at all. **Node voltage analysis** works on the original circuit, without changing it. It works for every LTI resistive circuit; voltage sources need small additions, such as the supernode. Its unknowns are only the $n-1$ node voltages; every branch voltage and current follows from them. Mesh current analysis comes later.

1. The procedure, on an example

For now, the circuit has only resistors and independent current sources; voltage sources and dependent sources come later. The circuit of Example 11.1 has $b = 7$ branches, so 14 branch voltages and currents. Node voltage analysis needs only 3 unknowns. Four steps find them.

Example 11.1 — node voltage analysis step by step



1. **Reference node.** Choose the bottom node as the ground; its node voltage is 0.
2. **Node voltages.** Label the others e_1, e_2, e_3. Every branch voltage is a difference of them ($v_{R_2} = e_1 - e_2$), so KVL holds automatically.
3. **Currents.** The current leaving node k through R to node j is $(e_k - e_j)/R$; to the ground, e_k/R.
4. **KCL** at every node except the ground:

$$\text{node 1:} \quad -i_{s1} + \frac{e_1}{R_1} + \frac{e_1 - e_2}{R_2} + \frac{e_1 - e_3}{R_5} = 0$$

$$\text{node 2:} \quad \frac{e_2 - e_1}{R_2} + \frac{e_2}{R_3} + \frac{e_2 - e_3}{R_4} = 0$$

$$\text{node 3:} \quad \frac{e_3 - e_2}{R_4} + \frac{e_3 - e_1}{R_5} + i_{s2} = 0$$

The source i_{s1} points into node 1, so it enters with a minus sign. Three equations, three unknowns; then Steps 2 and 3 give every branch voltage and current.

In general, $\mathbf{v} = A^T\mathbf{e}$ (KVL holds automatically), the element laws give $\mathbf{i}$, and KCL $A\mathbf{i} = \mathbf{0}$ gives $n-1$ equations in the $n-1$ node voltages.

2. Matrix form

With $G_k = 1/R_k$ and the terms collected, the three equations of Example 11.1 become

$$\underbrace{\begin{bmatrix} G_1+G_2+G_5 & -G_2 & -G_5 \\ -G_2 & G_2+G_3+G_4 & -G_4 \\ -G_5 & -G_4 & G_4+G_5 \end{bmatrix}}_{Y_n} \underbrace{\begin{bmatrix} e_1 \\ e_2 \\ e_3 \end{bmatrix}}_{\mathbf{e}} = \underbrace{\begin{bmatrix} i_{s1} \\ 0 \\ -i_{s2} \end{bmatrix}}_{\mathbf{i}_s}.$$

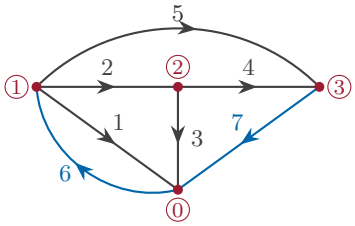
Y_n is the **node admittance matrix**, and $Y_n\mathbf{e} = \mathbf{i}_s$ are the **node equations**. For a circuit of resistors and current sources, they can be written down directly.

Writing Y_n directly from the circuit: three rules

- **Diagonal** entry (k,k): the sum of the conductances connected to node k.
- **Off-diagonal** entry (k,j): minus the sum of the conductances between nodes k and j.
- **Right side**, row k: the sum of the source currents entering node k.

So Y_n is **symmetric**: the entry (k,j) equals the entry (j,k).

Y_n from the graph. In the graph of Example 11.1, branches 1–5 are the resistors $R_1,\dots,R_5$, branches 6 and 7 are the sources i_{s1}, i_{s2}, and node ⓪ is the ground. The reduced incidence matrix A has one row for each node except the ground; an entry is $+1$ if the branch leaves the node and -1 if it enters.



$$A = \begin{array}{c} \\ ① \\ ② \\ ③ \end{array}\!\!\begin{array}{c} \begin{array}{ccccccc} 1 & 2 & 3 & 4 & 5 & 6 & 7 \end{array} \\ \left[\begin{array}{ccccc|cc} 1 & 1 & 0 & 0 & 1 & -1 & 0 \\ 0 & -1 & 1 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & -1 & -1 & 0 & 1 \end{array}\right] \end{array}$$

A_R: resistors A_S: sources

KCL $A\mathbf{i} = \mathbf{0}$ reads $A_R\mathbf{i}_R + A_S\mathbf{i}_S = \mathbf{0}$. The resistors obey $\mathbf{i}_R = G\,\mathbf{v}_R$ with $G = \mathrm{diag}(G_1,\dots,G_5)$. KVL, $\mathbf{v} = A^T\mathbf{e}$, splits the same way into $\mathbf{v}_R = A_R^T\mathbf{e}$ and $\mathbf{v}_S = A_S^T\mathbf{e}$; only the resistor rows are needed. In Example 11.1, A_R^T is the resistor columns 1–5 of A written as rows, and

$$\mathbf{v}_R = A_R^T\mathbf{e} = \begin{bmatrix} 1 & 1 & 0 & 0 & 1 \\ 0 & -1 & 1 & 1 & 0 \\ 0 & 0 & 0 & -1 & -1 \end{bmatrix}^T \begin{bmatrix} e_1 \\ e_2 \\ e_3 \end{bmatrix} = \begin{bmatrix} e_1 & e_1 - e_2 & e_2 & e_2 - e_3 & e_1 - e_3 \end{bmatrix}^T :$$

each resistor voltage is its tail node voltage minus its head node voltage. Together,

$$A_R\,G\,A_R^T\,\mathbf{e} = -A_S\mathbf{i}_S \quad \Rightarrow \quad Y_n = A_R\,G\,A_R^T, \qquad \mathbf{i}_s = -A_S\mathbf{i}_S.$$

For Example 11.1,

$$A_R\,G = \begin{bmatrix} G_1 & G_2 & 0 & 0 & G_5 \\ 0 & -G_2 & G_3 & G_4 & 0 \\ 0 & 0 & 0 & -G_4 & -G_5 \end{bmatrix}, \qquad -A_S\mathbf{i}_S = -\begin{bmatrix} -1 & 0 \\ 0 & 0 \\ 0 & 1 \end{bmatrix}\begin{bmatrix} i_{s1} \\ i_{s2} \end{bmatrix} = \begin{bmatrix} i_{s1} \\ 0 \\ -i_{s2} \end{bmatrix},$$

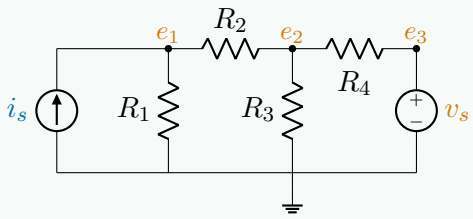
and $A_R\,G\,A_R^T$ is exactly the Y_n above: the three rules read this product off entry by entry. Finally, $Y_n^T = A_R\,G^T A_R^T = A_R\,G\,A_R^T = Y_n$, since G is diagonal: with resistors and current sources, Y_n is always symmetric.

3. A voltage source to the ground

A voltage source to the ground

If an independent voltage source v_s is connected between the ground and node k, then $e_k = v_s$ is known. So e_k is not an unknown, and KCL is not written at node k.

Example 11.2 — a voltage source between node 3 and the ground



The source fixes $e_3 = v_s$, so the unknowns are only e_1, e_2:

$$\text{node 1:} \quad -i_s + \frac{e_1}{R_1} + \frac{e_1 - e_2}{R_2} = 0$$

$$\text{node 2:} \quad \frac{e_2 - e_1}{R_2} + \frac{e_2}{R_3} + \frac{e_2 - v_s}{R_4} = 0$$

In matrix form, Example 11.2 reads

$$\begin{bmatrix} \frac{1}{R_1} + \frac{1}{R_2} & -\frac{1}{R_2} \\ -\frac{1}{R_2} & \frac{1}{R_2} + \frac{1}{R_3} + \frac{1}{R_4} \end{bmatrix} \begin{bmatrix} e_1 \\ e_2 \end{bmatrix} = \begin{bmatrix} i_s \\ v_s/R_4 \end{bmatrix}.$$

The known v_s moves to the right side: v_s/R_4 acts like a current into node 2.

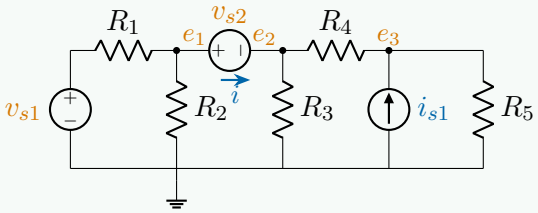
4. A voltage source between two nodes: modified node analysis

If a voltage source (independent or dependent) lies between two nodes and neither is the ground, KCL there needs its current. Its law fixes its voltage, not its current, so this current cannot be written with node voltages. **Modified node analysis** offers two ways out.

Method I

Let the unknown current i through the voltage source be one more unknown. Write KCL at every node except the ground, and add the law of the source as a **constraint equation**.

Example 11.3 — Method I



The top of v_{s1} is at the known voltage v_{s1}; v_{s2} is between nodes 1 and 2. Unknowns: e_1, e_2, e_3 and i.

$$\text{node 1:} \quad \frac{e_1 - v_{s1}}{R_1} + \frac{e_1}{R_2} + i = 0$$

$$\text{node 2:} \quad -i + \frac{e_2}{R_3} + \frac{e_2 - e_3}{R_4} = 0$$

$$\text{node 3:} \quad \frac{e_3 - e_2}{R_4} - i_{s1} + \frac{e_3}{R_5} = 0$$

$$\text{constraint:} \quad e_1 - e_2 = v_{s2}$$

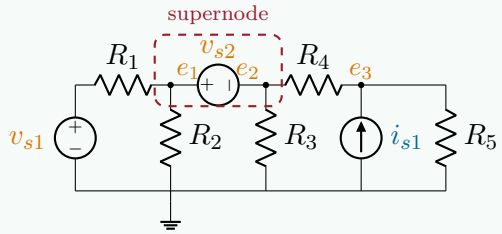
$$\begin{bmatrix} G_1 + G_2 & 0 & 0 & 1 \\ 0 & G_3 + G_4 & -G_4 & -1 \\ 0 & -G_4 & G_4 + G_5 & 0 \\ 1 & -1 & 0 & 0 \end{bmatrix} \begin{bmatrix} e_1 \\ e_2 \\ e_3 \\ i \end{bmatrix} = \begin{bmatrix} G_1 v_{s1} \\ 0 \\ i_{s1} \\ v_{s2} \end{bmatrix}$$

Four unknowns and four equations; the last row is the constraint. This works, but it adds the unknown i; the supernode below avoids it.

Method II: the supernode

Write KCL as if the two nodes of the voltage source were a single node, a **supernode**. Add the law of the source as a constraint equation.

Example 11.3, revisited — Method II, three unknowns



supernode: $\dfrac{e_1 - v_{s1}}{R_1} + \dfrac{e_1}{R_2} + \dfrac{e_2}{R_3} + \dfrac{e_2 - e_3}{R_4} = 0$

node 3: $\dfrac{e_3 - e_2}{R_4} - i_{s1} + \dfrac{e_3}{R_5} = 0$

constraint: $e_1 - e_2 = v_{s2}$

Why does it work? Add the KCL equations of nodes 1 and 2 in Method I: i leaves node 1 and enters node 2, so it cancels. The sum is KCL around the supernode. The current i is no longer needed. In matrix form, with three unknowns and three equations:

$$\begin{bmatrix} G_1+G_2 & G_3+G_4 & -G_4 \\ 0 & -G_4 & G_4+G_5 \\ 1 & -1 & 0 \end{bmatrix} \begin{bmatrix} e_1 \\ e_2 \\ e_3 \end{bmatrix} = \begin{bmatrix} G_1 v_{s1} \\ i_{s1} \\ v_{s2} \end{bmatrix}.$$

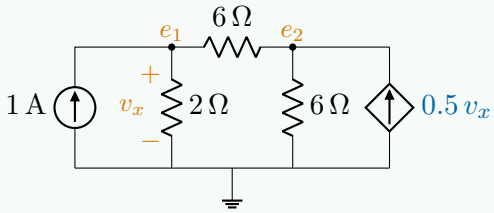
The matrix is no longer symmetric.

Method II needs one unknown fewer than Method I and gives the same node voltages; the current i, if needed, follows from a KCL afterwards.

5. Dependent sources

A dependent source is treated like an independent source of the same kind. Its control variable is written with node voltages; this gives one more constraint equation. A dependent source makes Y_n not symmetric.

Example 11.4 — a dependent current source



Question. Find e_1 and e_2. Treat the dependent source as a current source $0.5\,v_x$; write its control v_x with node voltages:

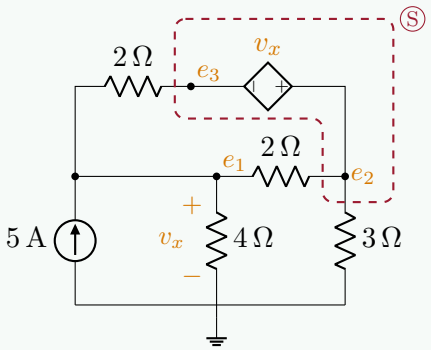
node 1: $-1 + \dfrac{e_1}{2} + \dfrac{e_1 - e_2}{6} = 0$

node 2: $\dfrac{e_2 - e_1}{6} + \dfrac{e_2}{6} - 0.5\,v_x = 0$

With $v_x = e_1$, the term $0.5\,e_1$ moves to the left side:

$$\begin{bmatrix} \frac{2}{3} & -\frac{1}{6} \\ -\frac{1}{6} - \frac{1}{2} & \frac{1}{3} \end{bmatrix} \begin{bmatrix} e_1 \\ e_2 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \quad e_1 = 3\,\text{V},\ e_2 = 6\,\text{V}.$$

Example 11.5 — a dependent voltage source



Question. Find the power absorbed by the 3 Ω resistor.

The source v_x is between nodes 3 and 2, and neither is the ground. So use a supernode Ⓢ around nodes 2 and 3. The unknowns are e_1, e_2, e_3.

(1) node 1: $-5 + \dfrac{e_1 - e_3}{2} + \dfrac{e_1}{4} + \dfrac{e_1 - e_2}{2} = 0$

(2) Ⓢ: $\dfrac{e_3 - e_1}{2} + \dfrac{e_2 - e_1}{2} + \dfrac{e_2}{3} = 0$

Constraint. The source gives $e_2 - e_3 = v_x$, and its control v_x is the voltage of the 4 Ω, so $v_x = e_1$:

$$e_2 - e_3 = e_1. \qquad (3)$$

Solving Example 11.5. Collect the terms of (1), (2) and (3), written as $e_1 - e_2 + e_3 = 0$:

$$\begin{bmatrix} \frac{5}{4} & -\frac{1}{2} & -\frac{1}{2} \\ -1 & \frac{5}{6} & \frac{1}{2} \\ 1 & -1 & 1 \end{bmatrix} \begin{bmatrix} e_1 \\ e_2 \\ e_3 \end{bmatrix} = \begin{bmatrix} 5 \\ 0 \\ 0 \end{bmatrix} \quad \Rightarrow \quad e_1 = 8\,\text{V},\ e_2 = 9\,\text{V},\ e_3 = 1\,\text{V}.$$

The $3\,\Omega$ is between node 2 and the ground:

$$P_{3\Omega} = \frac{{e_2}^2}{3} = \frac{9^2}{3} = 27\,\text{W}.$$

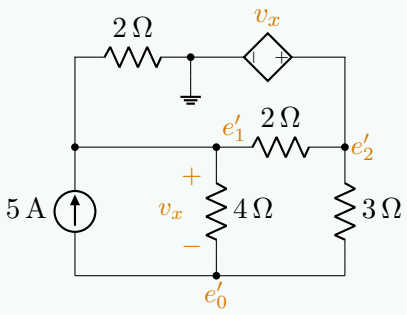
Remark: a dependent voltage source to the ground

If a dependent voltage source is connected between node k and the ground, do not write KCL at node k: its current is not known. Write the constraint that the dependent source gives instead.

6. Choosing a different ground

Any node can be the ground. Changing it shifts every node voltage by the same amount, so the branch voltages, the currents and the powers do not change.

Example 11.6 — Example 11.5 with node 3 as the ground



Question. Take node 3 as the ground. Does $P_{3\Omega}$ change?

Answer. Every node voltage drops by the same $e_3 = 1\,\text{V}$: $e'_1 = 7\,\text{V}$, $e'_2 = 8\,\text{V}$, $e'_0 = -1\,\text{V}$. Branch voltages are differences, so they do not change:

$$P_{3\Omega} = \frac{(e'_2 - e'_0)^2}{3} = \frac{9^2}{3} = 27\,\text{W}, \quad \text{the same.}$$

With this ground, v_x is between node 2 and the ground: no supernode is needed.

Check in matrix form. KCL at nodes 1 and 0, and the constraint $e'_2 = v_x = e'_1 - e'_0$:

$$\begin{bmatrix} \frac{5}{4} & -\frac{1}{2} & -\frac{1}{4} \\ -\frac{1}{4} & -\frac{1}{3} & \frac{7}{12} \\ 1 & -1 & -1 \end{bmatrix} \begin{bmatrix} e'_1 \\ e'_2 \\ e'_0 \end{bmatrix} = \begin{bmatrix} 5 \\ -5 \\ 0 \end{bmatrix} \quad \Rightarrow \quad e'_1 = 7\,\text{V},\ e'_2 = 8\,\text{V},\ e'_0 = -1\,\text{V}.$$

Tip. Put the ground at one end of a voltage source. Then its other node voltage is known, or a supernode is avoided.

Summary

In the circuit	What to do
resistors and current sources	KCL at every non-ground node: $Y_n\mathbf{e} = \mathbf{i}_s$, from three simple rules; $Y_n = A_R G A_R^T$ is symmetric
a voltage source from the ground to node k	$e_k = v_s$ is known; no KCL at node k
a voltage source between two nodes	Method I: its current as an unknown, plus the constraint; Method II: KCL at the supernode, plus the constraint
a dependent source	write its control variable with node voltages: a constraint equation
the choice of ground	any node; the branch voltages and currents do not change

After the node voltages are found: $\mathbf{v} = A^T\mathbf{e}$, then the branch currents from the element laws.